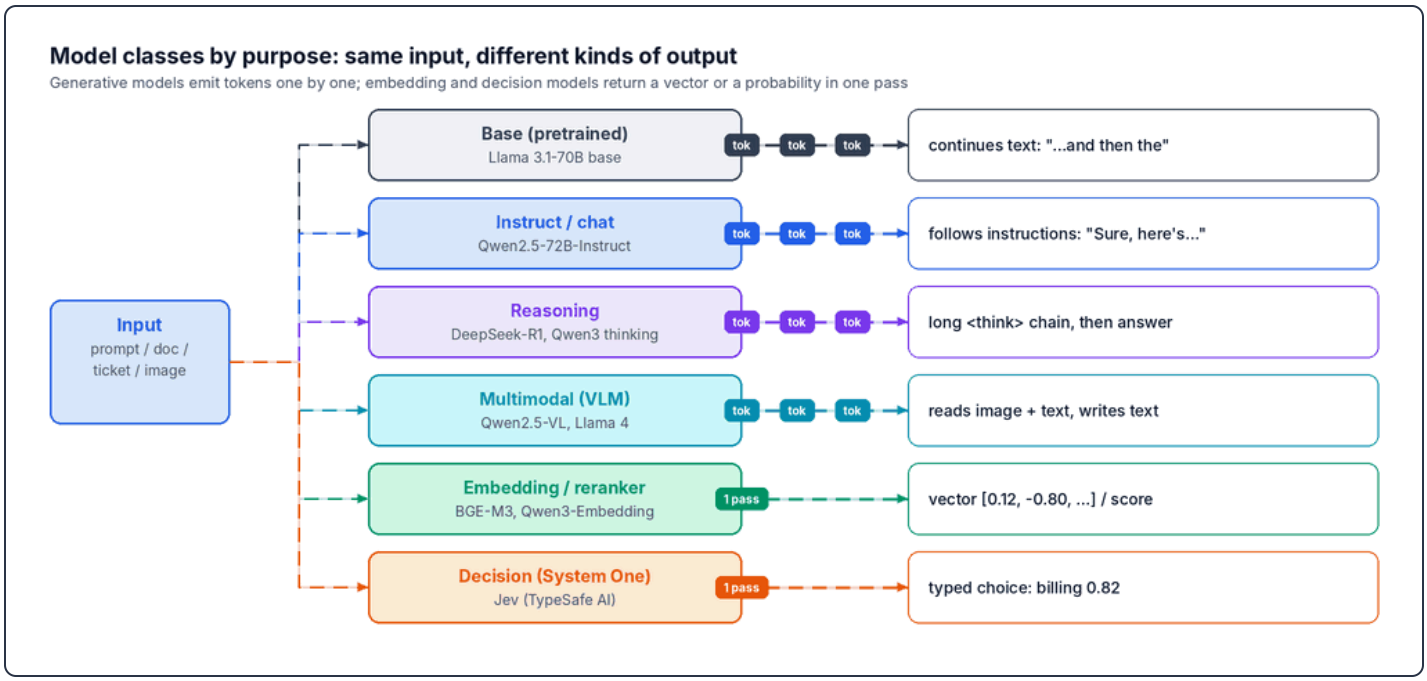
LLM inference diagrams

The 11 animated diagrams from the architect's guide. Each one loops continuously. On a phone, scroll a diagram sideways to see all of it.

Model classes Number formats and quantization Dense vs MoE vs Hybrid Attention variants
Transformer block and SwiGLU FFN KV cache design FLOPs: prefill vs decode VRAM and weight loading
Claude tiers Self-hosted prefill/decode stack Claude on Bedrock for 1,000+ users

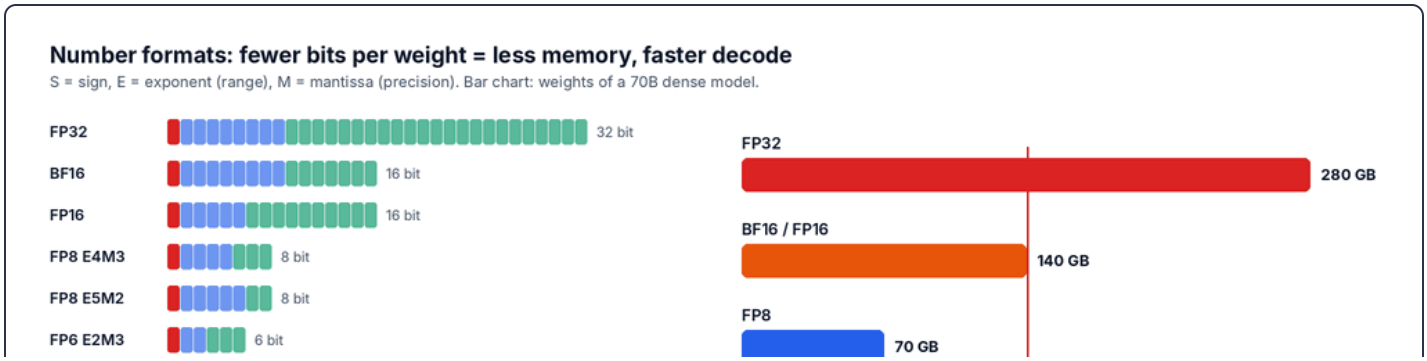
1. Model classes

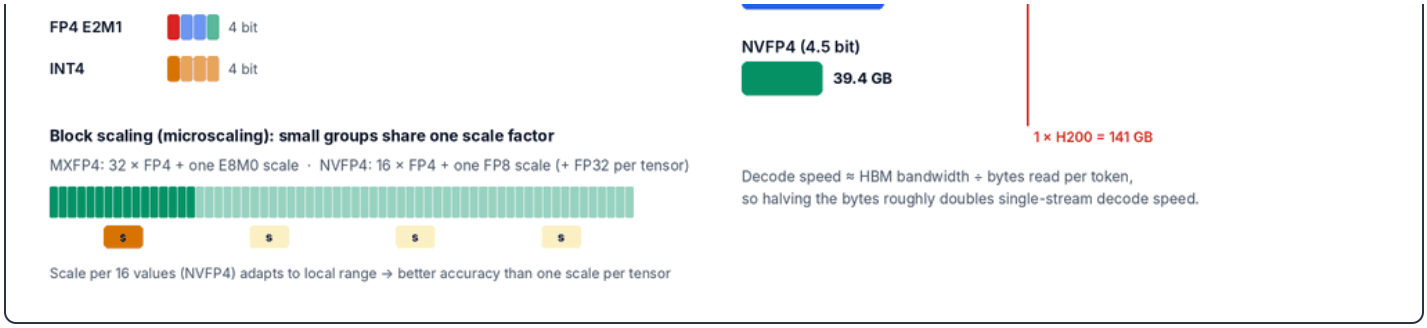
Generative classes stream tokens one at a time; embedding and decision models (like Jev) answer in a single pass.



2. Number formats and quantization

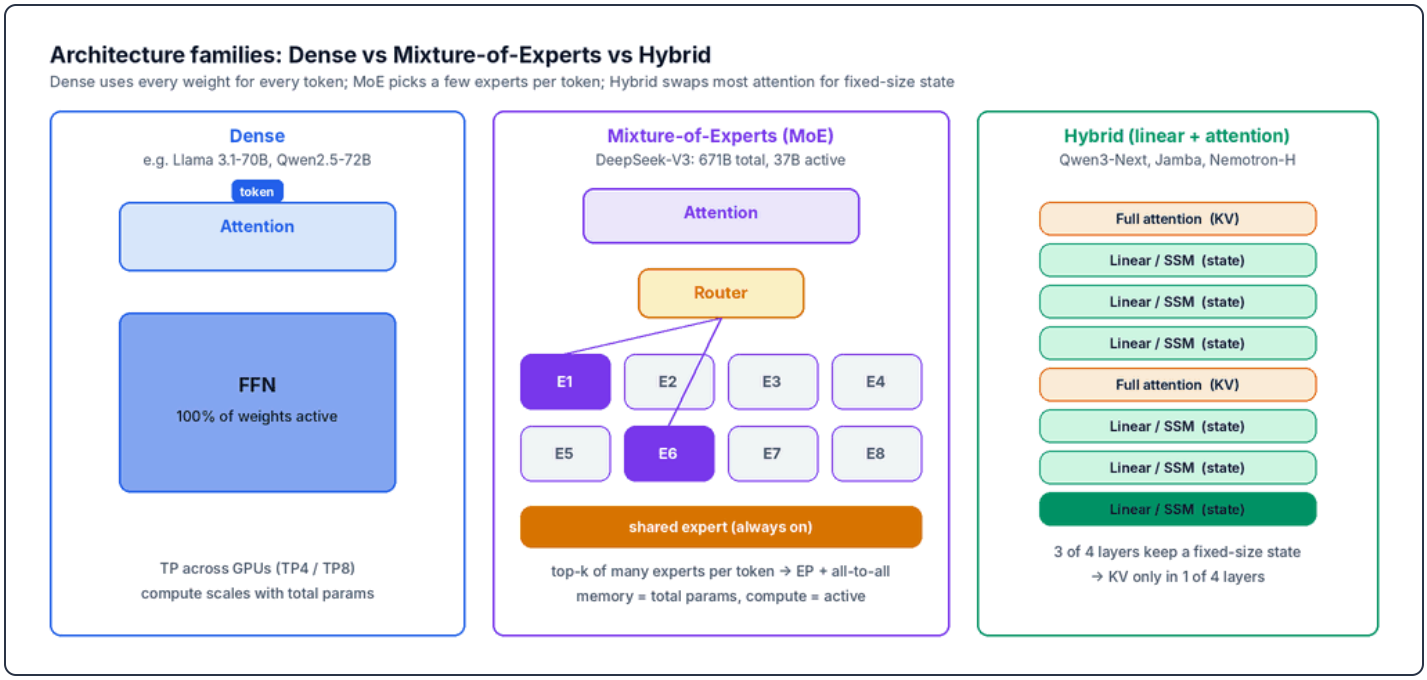
Bit layouts sweep left to right, NVFP4 scale groups light up in turn, and the 70B weight-memory bars grow against one H200's 141 GB.





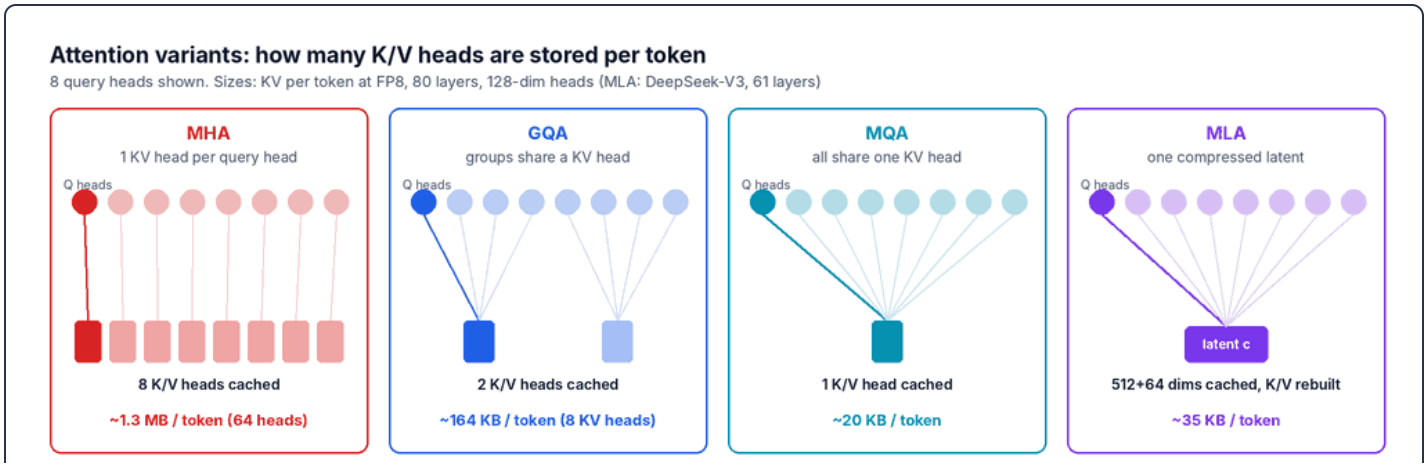
3. Dense vs MoE vs Hybrid

Dense lights the whole FFN; MoE's router picks two new experts for each token; Hybrid steps through linear and attention layers.



4. Attention variants

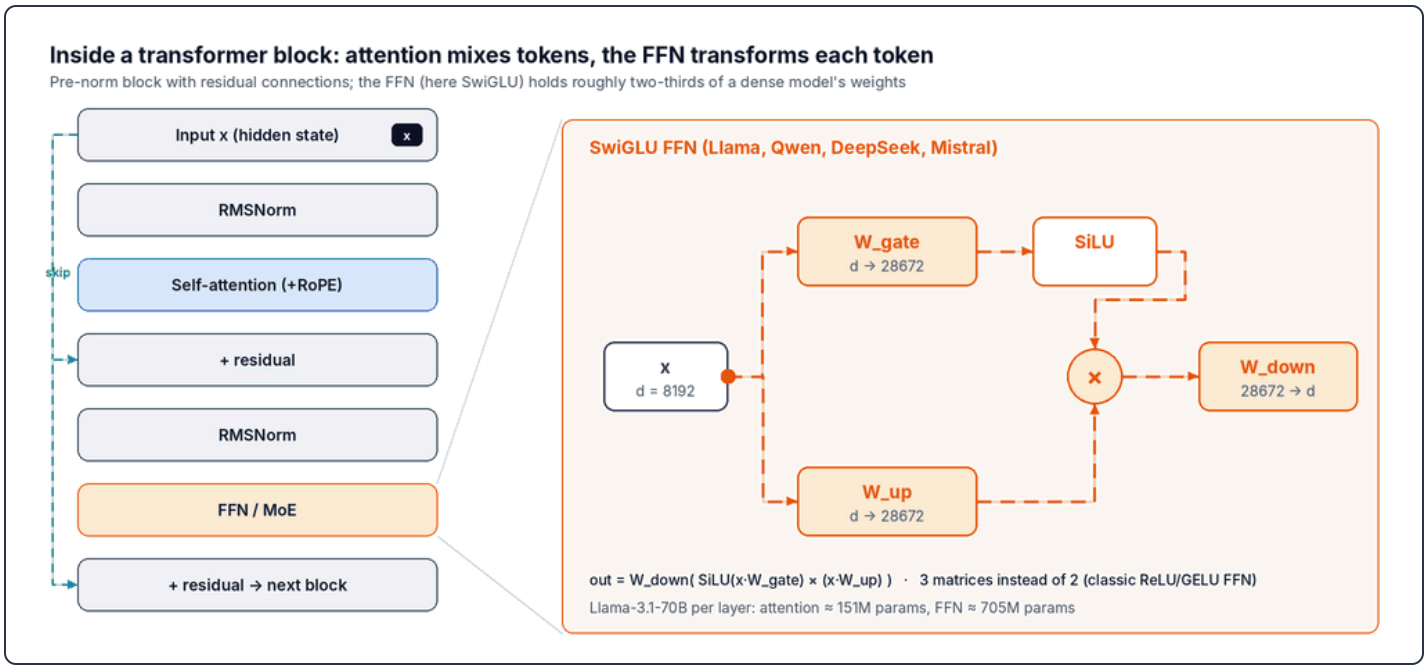
Each query head lights up with the K/V head it reads. Bottom: full causal attention keeps every past token, sliding window keeps a fixed window.





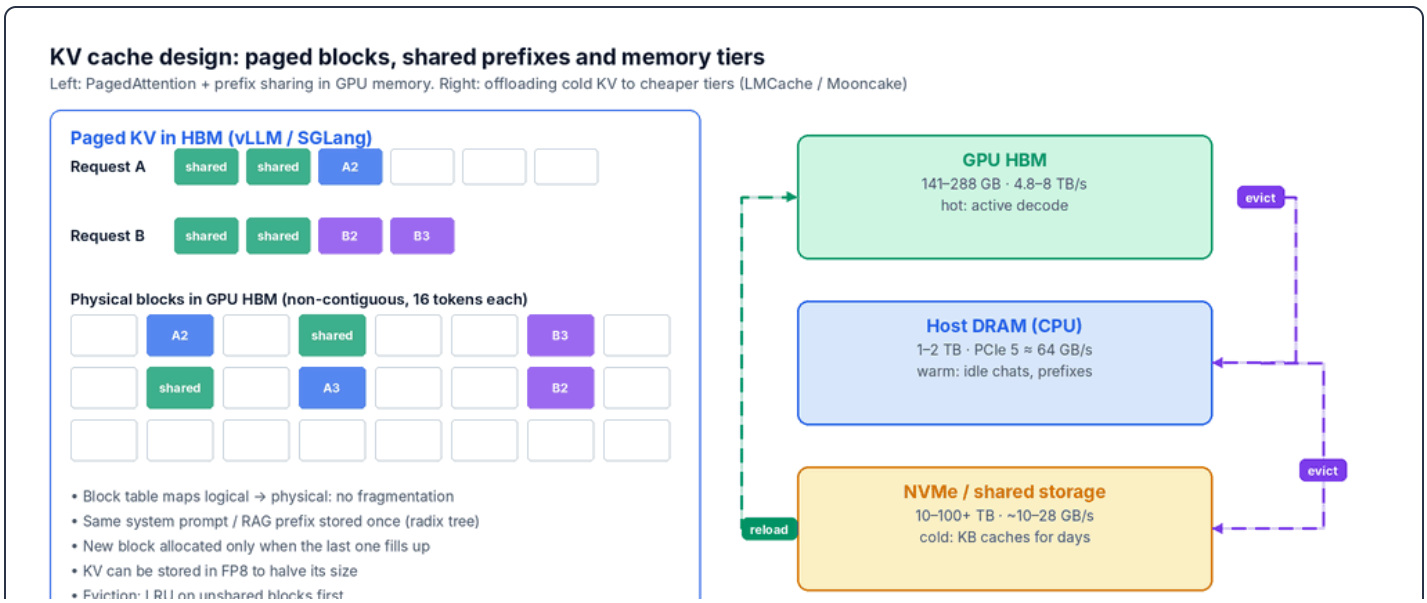
5. Transformer block and SwiGLU FFN

A token moves through the block with residual skips; data flows through the gate, up and down projections.



6. KV cache design

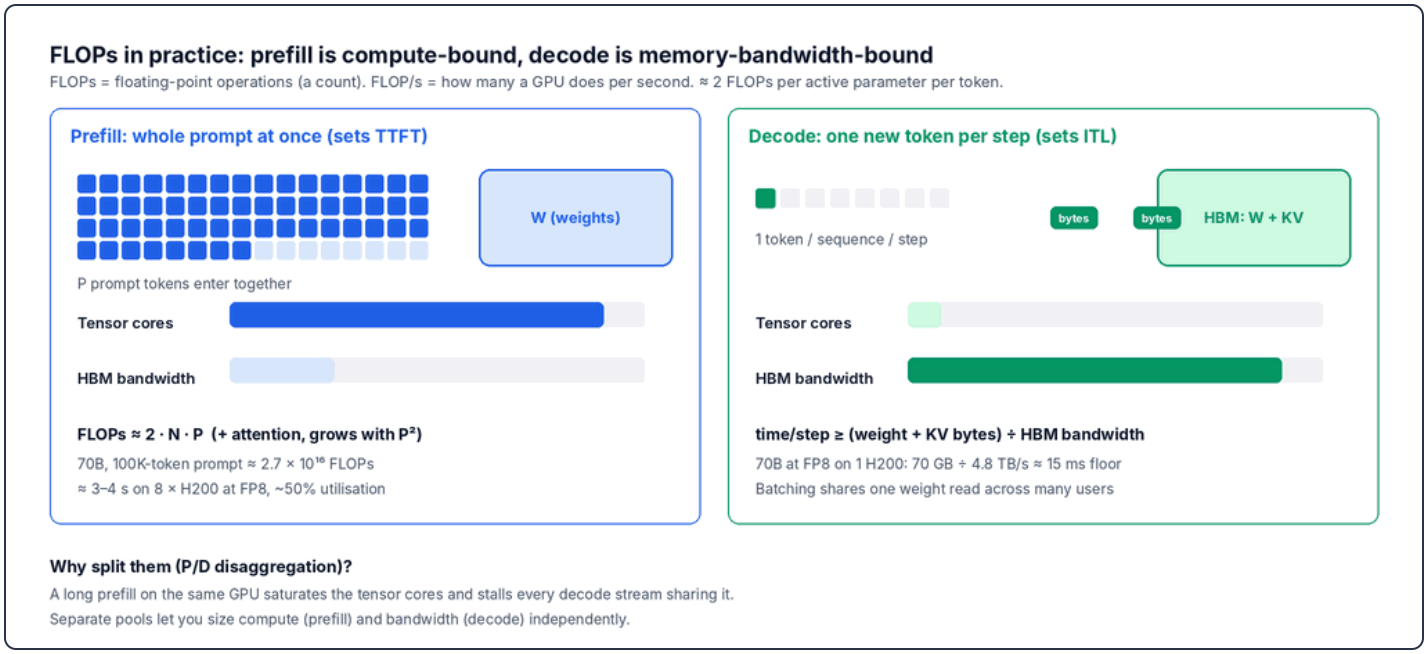
Request A fills new paged blocks while sharing a prefix with B; cold KV is evicted down the tiers and reloaded on a hit.



Reloading cached KV beats recomputing prefill when the link is fast enough

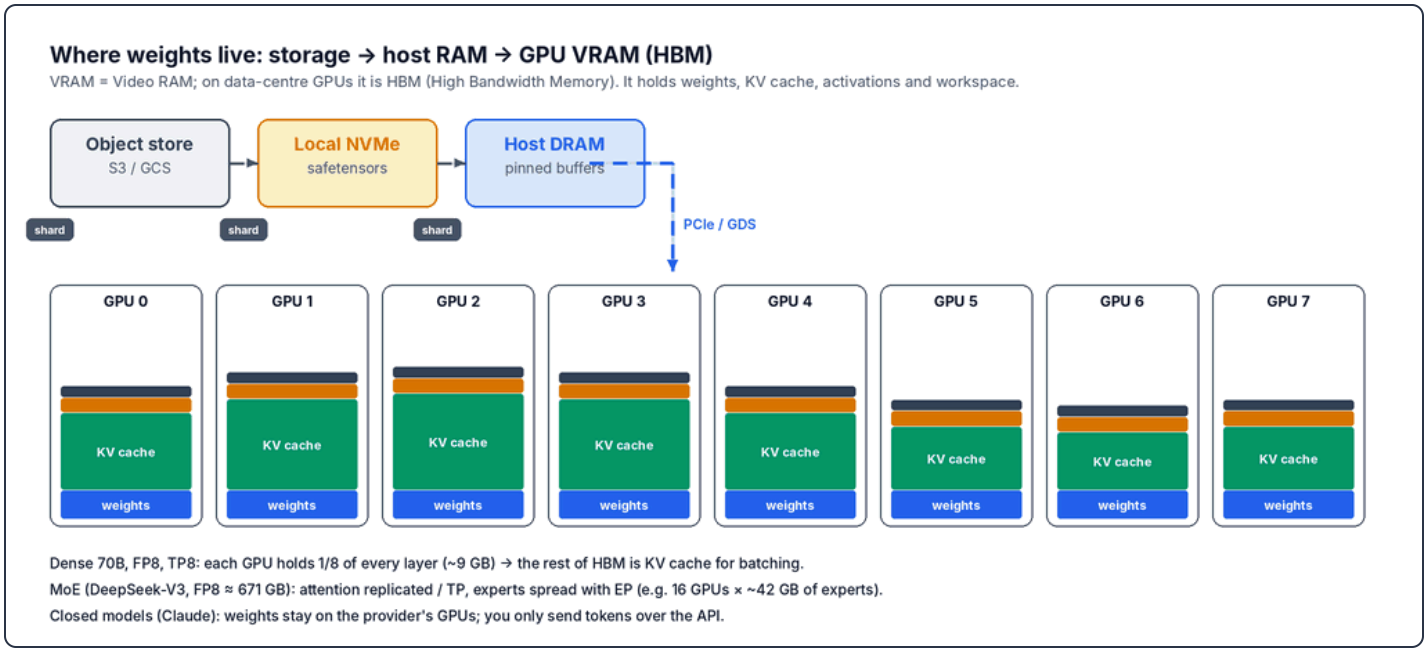
7. FLOPs: prefill vs decode

Prefill keeps tensor cores busy with many tokens at once; decode adds one token per step and streams weight bytes from HBM.



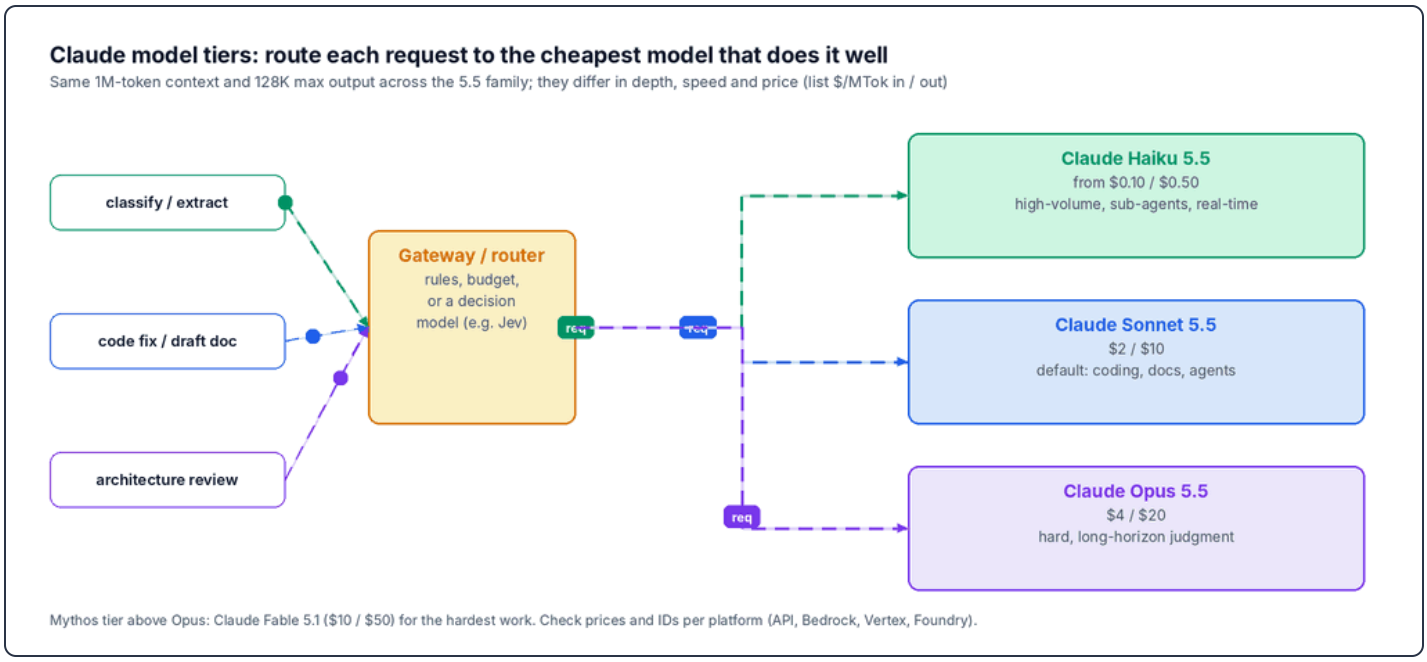
8. VRAM and weight loading

Shards stream from storage into 8 GPUs; the KV cache share of each GPU's memory rises and falls with load.



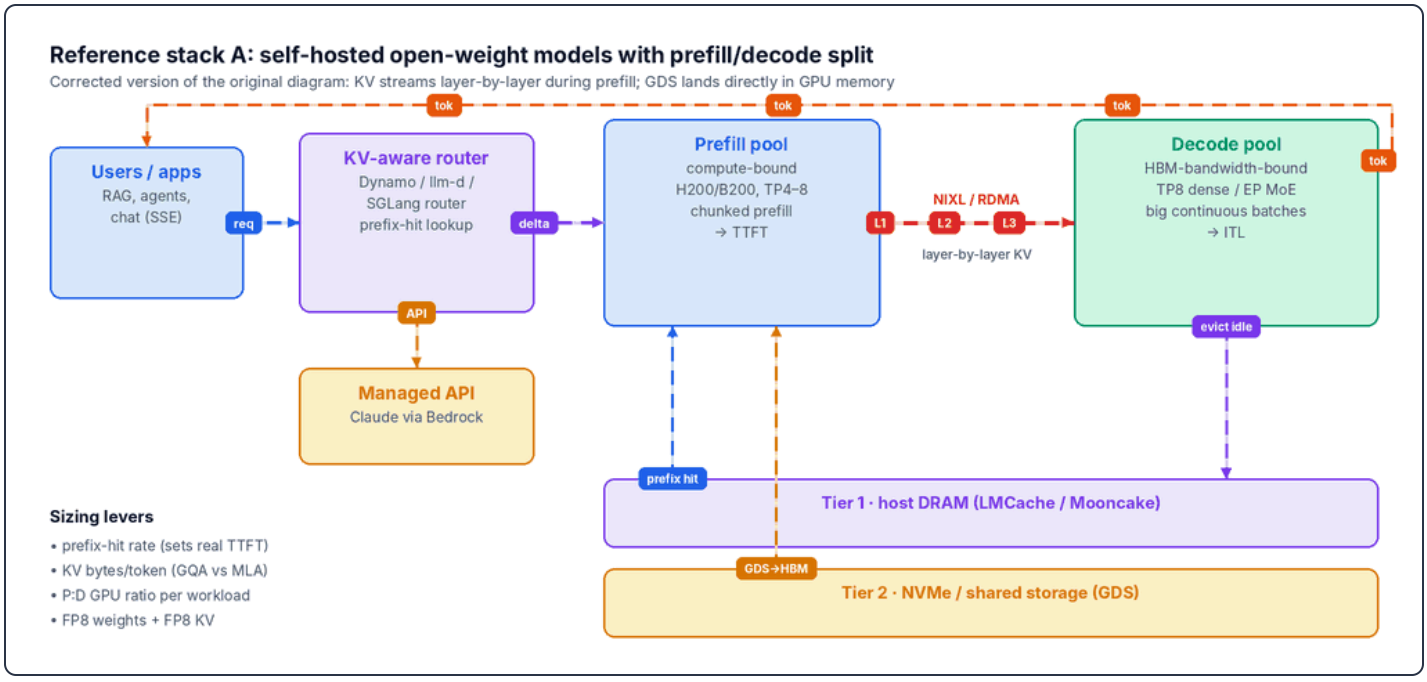
9. Claude tiers

The gateway routes requests to Haiku, Sonnet or Opus; Haiku requests move fastest.



10. Self-hosted prefill/decode stack

Requests go through the router to prefill, KV streams layer by layer to decode, and tokens flow back to users.



11. Claude on Bedrock for 1,000+ users

Clients pass SSO and the edge, the gateway calls Bedrock over PrivateLink, and responses stream back.

Reference stack B: Claude on Amazon Bedrock for 1,000+ enterprise users

No GPUs to run: you operate the gateway (identity, quotas, routing, logging); AWS runs inference

